

CLOUD, MICROSERVICES, & APPLICATIONS

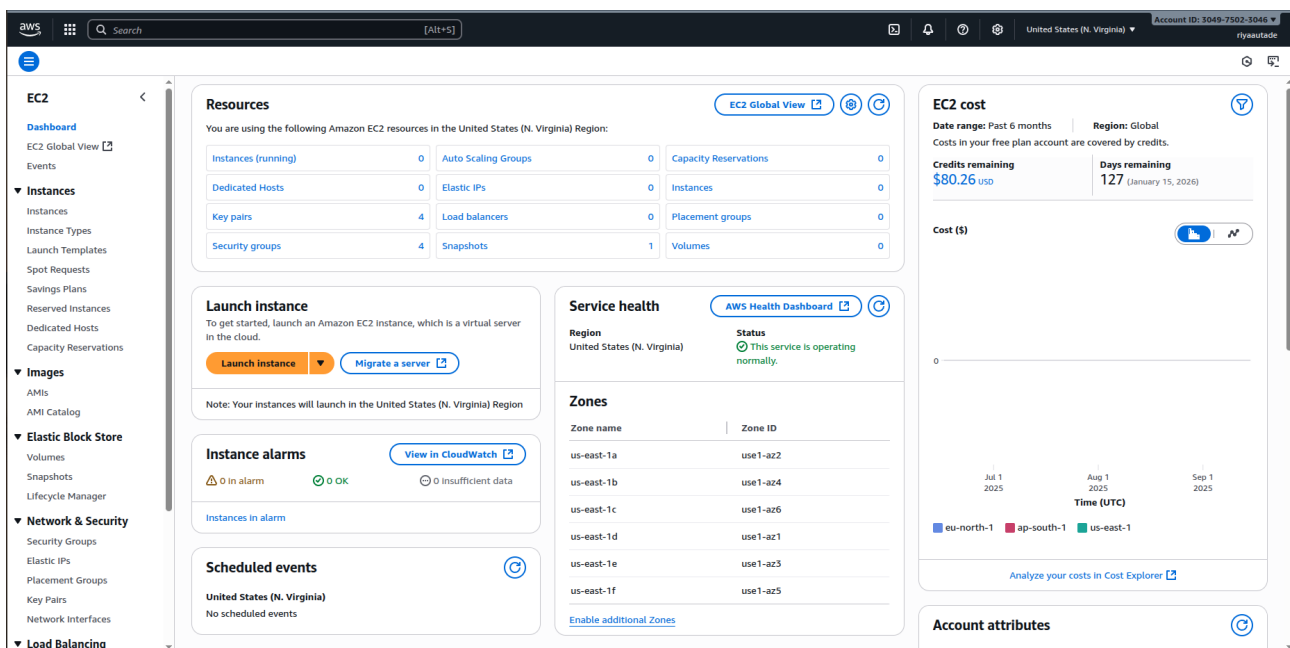
MIDTERM EXAMINATION

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22BBS0079

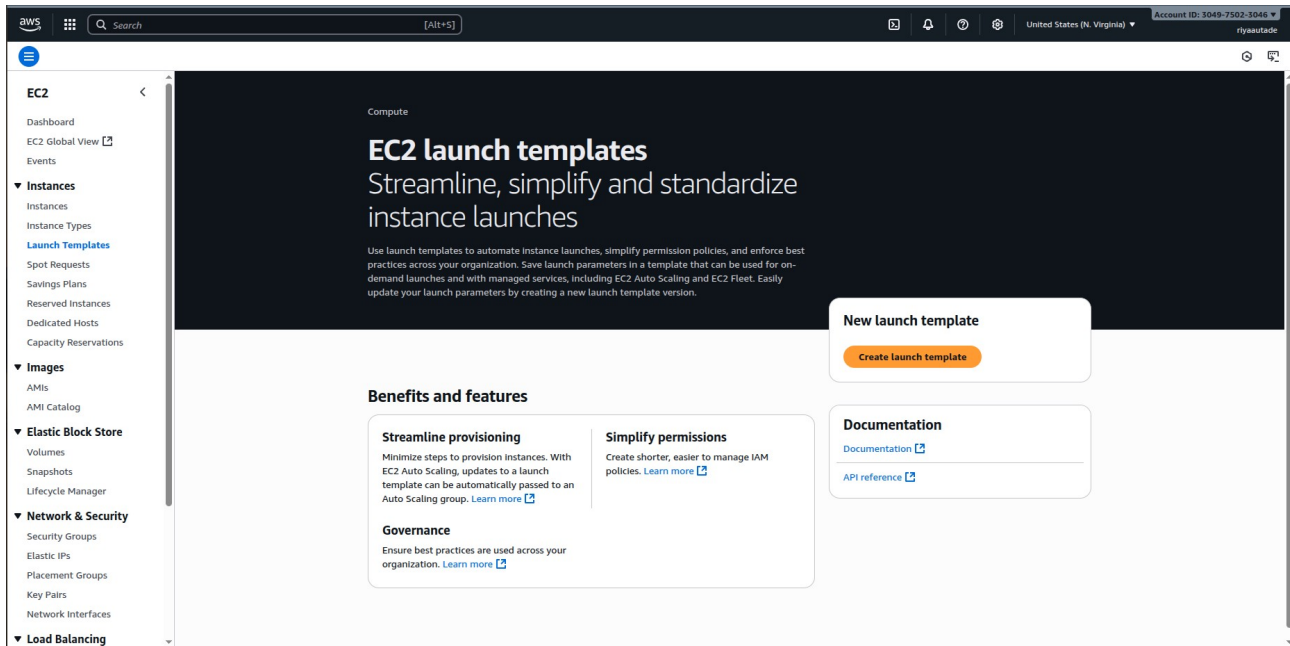
Question:

Implement EC2 Auto Scaling with a Launch Template. Create a Key Pair and a Security Group allowing HTTP (port 80) and SSH (port 22) access. Create a Launch Template using Amazon Linux 2 with a user-data script to install Apache on startup. Create an Auto Scaling Group attached to the default VPC with a minimum of one instance and a maximum of three instances. Configure scaling policies to scale out when CPU utilization exceeds 50% and scale in when it drops below 20%. Test the scaling behavior by generating CPU load and verify Apache access. Document all steps with screenshots.

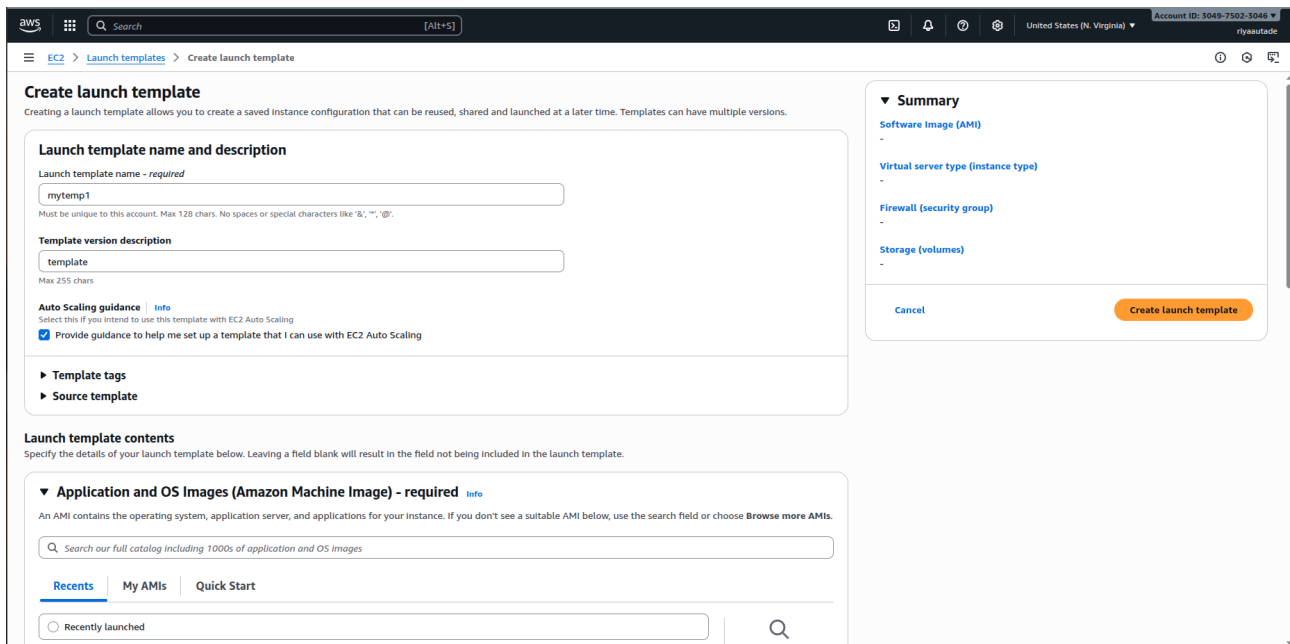
1. Go to EC2 Dashboard and click on Launch Templates:



2. Click on Create Launch template



3. Name the template, give a description and choose “Provide Guidance” option.



4. Go to Quick Start→ Amazon Linux under AMI. Then choose t3.micro instance

Application and OS images (Amazon Machine Image) - required

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Recents | **My AMIs** | **Quick Start**

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-00ca32bbc84273381 (64-bit x86), uefi-preferred / ami-0aa7db6294d00216f (64-bit ARM), uefi
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.8.20250818.0 x86_64 HVM kernel-6.1

Architecture 64-bit (x86) **Boot mode** uefi-preferred **AMI ID** ami-00ca32bbc84273381 **Publish Date** 2025-08-15 **Username** ec2-user **Verified provider**

Instance type

t3.micro 2 vCPU 1 GiB Memory Current generation: true On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand Linux base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0593 USD per Hour On-Demand Windows base pricing: 0.0186 USD per Hour

Summary

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...read more
ami-00ca32bbc84273381

Virtual server type (instance type)
t3.micro

Firewall (security group)
-

Storage (volumes)
1 volume(s) - 8 GiB

Create launch template

5. Create a key pair, name it, choose type RSA and .pem

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the key pair.

Key pair name
Don't include in launch template

Network settings

Subnet
Don't include in launch template

Availability Zone
Don't include in launch template

Firewall (security groups)
Select existing security group

Security groups
Select security groups

Storage (volumes)
EBS Volumes

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.
it

Key pair type
☒ RSA
RSA encrypted private and public key pair
☐ ED25519
ED25519 encrypted private and public key pair

Private key file format
☒ .pem
For use with OpenSSH
☐ .ppk
For use with PuTTY

When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Create key pair

6. Make a new security group with ports 80 and 22 allowed

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-/!@,[]+=&; {}\$*

Description - required | Info
Allows SSH access to developers

VPC | Info
vpc-052c2b25aadc0d07cd (default) 172.31.0.0/16

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 80, 0.0.0.0/0) Remove

Type | Info: HTTP
Protocol | Info: TCP
Port range | Info: 80
Source type | Info: Anywhere
Source | Info: 0.0.0.0/0
Description - optional | Info: e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 22, 0.0.0.0/0) Remove

Type | Info: ssh
Protocol | Info: TCP
Port range | Info: 22
Source type | Info: Anywhere
Source | Info: 0.0.0.0/0
Description - optional | Info: e.g. SSH for admin desktop

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ×

Add security group rule

Summary

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-00ca32bbc84273381

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Create launch template

7. Enable auto assign public IP, and make delete on termination 'Yes'

Advanced network configuration

Network interface 1 Remove

Device index | Info: 0

Subnet | Info: Don't include in launch template
Not applicable for EC2 Auto Scaling

Primary IP | Info: Don't include in launch template
Not applicable for EC2 Auto Scaling

IPv4 Prefixes | Info: Don't include in launch template

Delete on termination | Info: Yes

ENA Express | Info: Don't include in launch template
The selected instance type does not support ENA Express.

Idle connection tracking timeout | Info: ☐ Enable

Network interface | Info: New Interface
Existing network interfaces are not recommended when creating a template for auto-scaling.

Security groups | Info: New security group

Secondary IP | Info: Don't include in launch template
Not applicable for EC2 Auto Scaling

IPv6 Prefixes | Info: Don't include in launch template

Interface type | Info: Don't include in launch template

ENA Express UDP | Info: Don't include in launch template
The selected instance type does not support ENA Express.

Auto-assign public IP | Info: Enable

IPv6 IPs | Info: Don't include in launch template
Not applicable for EC2 Auto Scaling

Assign Primary IPv6 IP | Info: Don't include in launch template

Network card index | Info: Don't include in launch template
The selected instance type does not support multiple network cards.

ENA queues | Info: The selected instance type does not support ENA queues.

Add network interface

Storage (volumes) | Info

Summary

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...[read more](#)
ami-00ca32bbc84273381

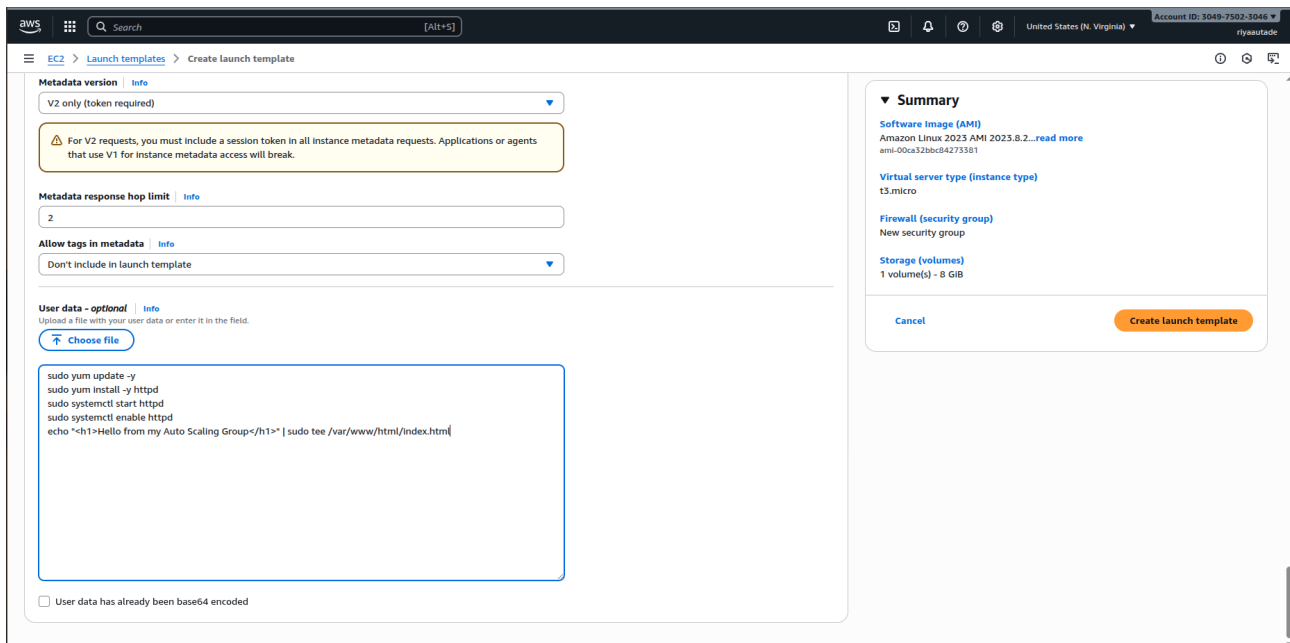
Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Create launch template

8. Add the user script for Apache



The screenshot shows the AWS Management Console interface for creating a launch template. The 'User data - optional' field is populated with the following script:

```
sudo yum update -y
sudo yum install -y httpd
sudo systemctl start httpd
sudo systemctl enable httpd
echo "<h1>Hello from my Auto Scaling Group</h1>" | sudo tee /var/www/html/index.html
```

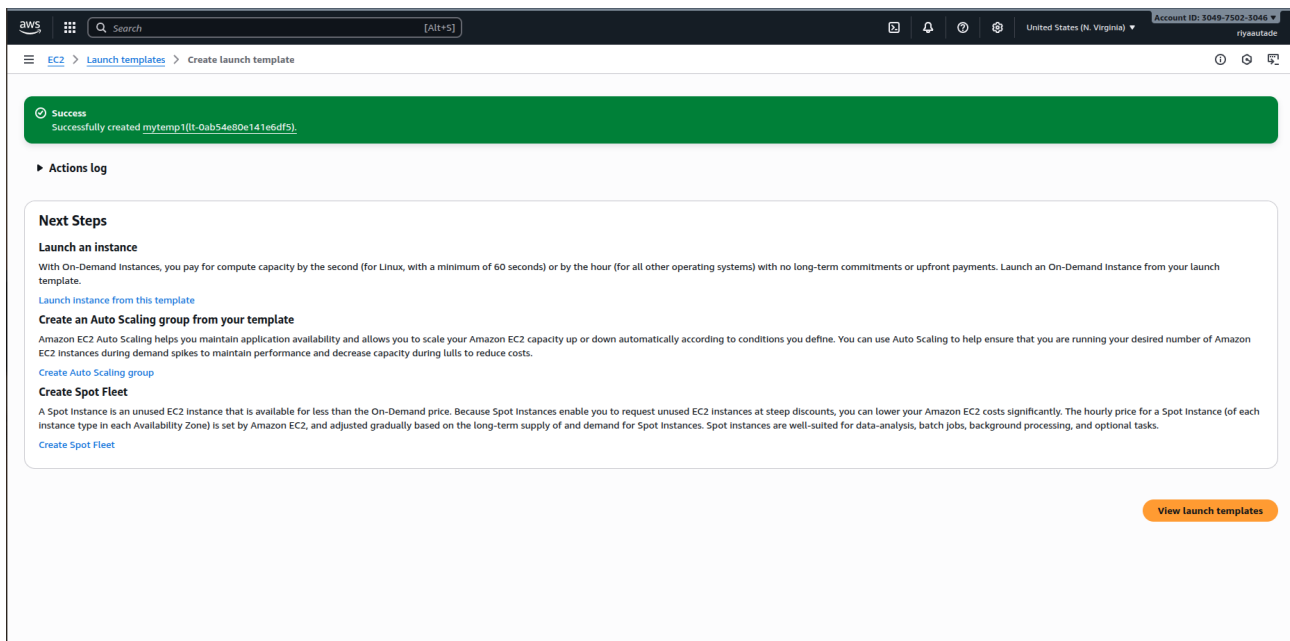
The 'Summary' panel on the right displays the following configuration:

- Software Image (AMI):** Amazon Linux 2023 AMI 2023.8.2...[read more](#) (ami-00ca32bbc84273381)
- Virtual server type (instance type):** t3.micro
- Firewall (security group):** New security group
- Storage (volumes):** 1 volume(s) - 8 GiB

Buttons for 'Cancel' and 'Create launch template' are visible at the bottom right of the summary panel.

```
sudo yum update -y
sudo yum install -y httpd
sudo systemctl start httpd
sudo systemctl enable httpd
echo "<h1>Hello from my Auto Scaling Group</h1>" | sudo tee /var/www/html/index.html
```

9. Click on create template.



The screenshot shows the AWS Management Console interface after successfully creating a launch template. A green success banner at the top reads: "Success Successfully created mytemp1ltt-0ab54e80e141e6df5)".

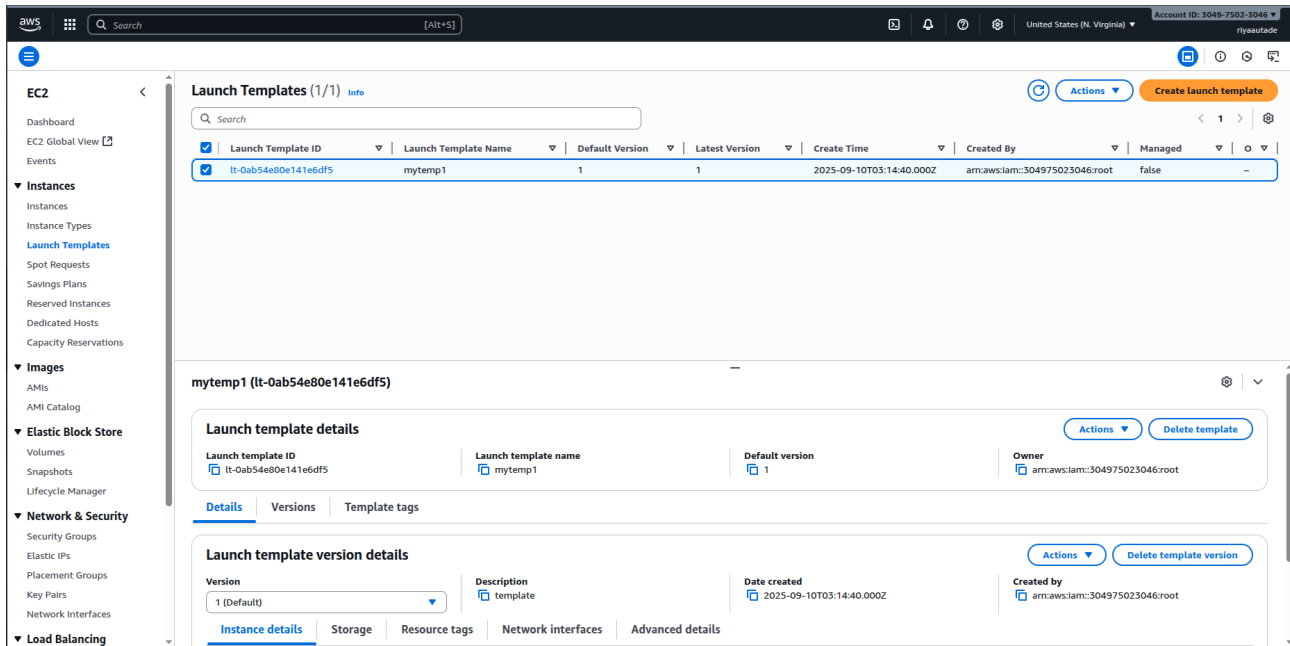
Below the banner is the 'Actions log' section, which is currently empty.

The 'Next Steps' section provides guidance on how to use the template:

- Launch an instance:** With On-Demand Instances, you pay for compute capacity by the second (for Linux, with a minimum of 60 seconds) or by the hour (for all other operating systems) with no long-term commitments or upfront payments. Launch an On-Demand Instance from your launch template. [Launch instance from this template](#)
- Create an Auto Scaling group from your template:** Amazon EC2 Auto Scaling helps you maintain application availability and allows you to scale your Amazon EC2 capacity up or down automatically according to conditions you define. You can use Auto Scaling to help ensure that you are running your desired number of Amazon EC2 instances during demand spikes to maintain performance and decrease capacity during lulls to reduce costs. [Create Auto Scaling group](#)
- Create Spot Fleet:** A Spot Instance is an unused EC2 instance that is available for less than the On-Demand price. Because Spot instances enable you to request unused EC2 instances at steep discounts, you can lower your Amazon EC2 costs significantly. The hourly price for a Spot instance (of each instance type in each Availability Zone) is set by Amazon EC2, and adjusted gradually based on the long-term supply of and demand for Spot instances. Spot instances are well-suited for data-analysis, batch jobs, background processing, and optional tasks. [Create Spot Fleet](#)

A 'View launch templates' button is located at the bottom right of the page.

10. view launch template and click on it



The screenshot shows the AWS Management Console interface for the 'Launch Templates' page. The left sidebar contains navigation links for EC2, Instances, Images, Elastic Block Store, Network & Security, and Load Balancing. The main content area displays a table of launch templates with columns for Launch Template ID, Launch Template Name, Default Version, Latest Version, Create Time, Created By, and Managed. The 'mytemp1' launch template is selected. Below the table, the 'Launch template details' section shows the template ID, name, and version. The 'Launch template version details' section shows the version number and description.

Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Created By	Managed
lt-0ab54e80e141e6df5	mytemp1	1	1	2025-09-10T03:14:40.000Z	arn:aws:iam::304975023046:root	false

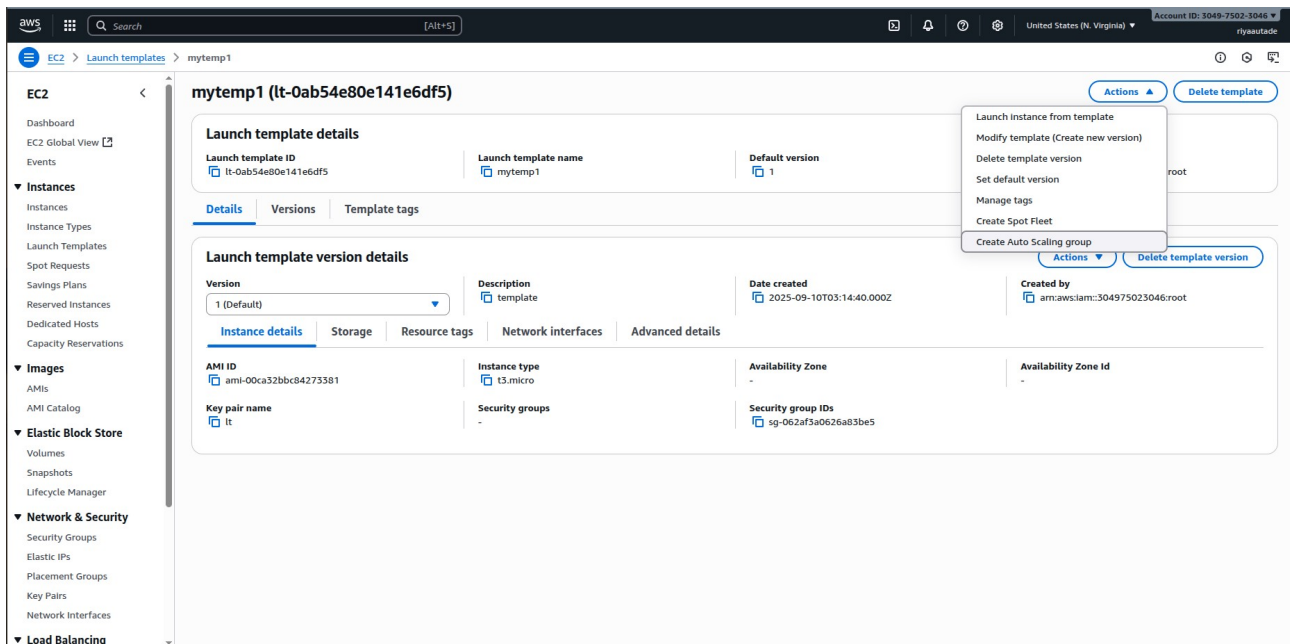
Launch template details

Launch template ID: lt-0ab54e80e141e6df5
Launch template name: mytemp1
Default version: 1
Owner: arn:aws:iam::304975023046:root

Launch template version details

Version: 1 (Default)
Description: template
Date created: 2025-09-10T03:14:40.000Z
Created by: arn:aws:iam::304975023046:root

11. Go to Actions → create Auto scaling group



The screenshot shows the AWS Management Console interface for the 'Launch Templates' page. The left sidebar contains navigation links for EC2, Instances, Images, Elastic Block Store, Network & Security, and Load Balancing. The main content area displays a table of launch templates with columns for Launch Template ID, Launch Template Name, Default Version, Latest Version, Create Time, Created By, and Managed. The 'mytemp1' launch template is selected. Below the table, the 'Launch template details' section shows the template ID, name, and version. The 'Launch template version details' section shows the version number and description. The 'Actions' dropdown menu is open, showing options like 'Launch instance from template', 'Modify template (Create new version)', 'Delete template version', 'Set default version', 'Manage tags', 'Create Spot Fleet', and 'Create Auto Scaling group'.

Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Created By	Managed
lt-0ab54e80e141e6df5	mytemp1	1	1	2025-09-10T03:14:40.000Z	arn:aws:iam::304975023046:root	false

Launch template details

Launch template ID: lt-0ab54e80e141e6df5
Launch template name: mytemp1
Default version: 1
Owner: arn:aws:iam::304975023046:root

Launch template version details

Version: 1 (Default)
Description: template
Date created: 2025-09-10T03:14:40.000Z
Created by: arn:aws:iam::304975023046:root

Instance details

AMI ID: ami-00ca32bbx84273381
Instance type: t3.micro
Availability Zone: -
Availability Zone Id: -
Key pair name: lt
Security groups: -
Security group IDs: sg-062af3a0626a83be5

12. name the template and choose the latest version

The screenshot shows the 'Choose launch template or configuration' step in the AWS Management Console. The left sidebar lists steps from 1 to 7, with step 1 'Choose launch template or configuration' selected. The main content area has a title 'Choose launch template or configuration' and a description. Below this, there are three main sections: 'Name', 'Launch template', and 'Additional details'.

Name

Auto Scaling group name
Enter a name to identify the group.

Must be unique to this account in the current Region and no more than 255 characters.

Launch template

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

[Create a launch template](#)

Version

[Create a launch template version](#)

Additional details

Property	Value
Description	template
Launch template	mytemp1 lt-Qab54e80e141e6df5
Instance type	t3.micro
AMI ID	ami-00ca32bbc84273381
Security groups	-
Request Spot Instances	No
Key pair name	lt
Security group IDs	sg-062af3a0626a83be5
Storage (volumes)	-
Date created	Wed Jan 10 20:25:08-04:00 GMT+05:30 (India Standard Time)

13. Add an availability zone like us-east-1a and click on Next

The screenshot shows the 'Instance type requirements' and 'Network' steps in the AWS Management Console. The left sidebar lists steps from 1 to 7, with step 2 'Choose instance launch options' selected. The main content area has a title 'Instance type requirements' and a description. Below this, there are three main sections: 'Instance type requirements', 'Network', and 'Availability Zone distribution'.

Instance type requirements

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Property	Value
Launch template	mytemp1 lt-Qab54e80e141e6df5
Version	Latest
Description	template
Instance type	t3.micro

[Override launch template](#)

Network

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

172.31.0.0/16 Default
[Create a VPC](#)

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

172.31.80.0/20 Default
[Create a subnet](#)

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

14. Configure group size and scaling as per the question:

Desired Capacity: 1

Minimum Desired Capacity: 1

Maximum Desired Capacity: 3

The screenshot shows the 'Create Auto Scaling group' wizard in the AWS Management Console, specifically Step 4: 'Configure group size and scaling'. The left sidebar shows the progress of the steps: Step 1 (Choose launch template or configuration), Step 2 (Choose instance launch options), Step 3 (optional, Integrate with other services), Step 4 (optional, Configure group size and scaling), Step 5 (optional, Add notifications), Step 6 (optional, Add tags), Step 7, and Review. Step 4 is currently selected.

Configure group size and scaling - optional [Info](#)

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances) ▼

Desired capacity

Specify your group size.

1

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

3

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

Target Tracking Policy

15. Click on Target Tracking Scaling Policy, name it, and click on Average CPU utilization with target value as 50 and instance warnings seconds as 300 (as per the question requirements)

The screenshot shows the 'Create Auto Scaling group' wizard in the AWS Management Console, specifically Step 4: 'Configure group size and scaling'. The left sidebar shows the progress of the steps: Step 1 (Choose launch template or configuration), Step 2 (Choose instance launch options), Step 3 (optional, Integrate with other services), Step 4 (optional, Configure group size and scaling), Step 5 (optional, Add notifications), Step 6 (optional, Add tags), Step 7, and Review. Step 4 is currently selected.

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

3

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name

Target Tracking Policy

Metric type [Info](#)

Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization ▼

Target value

50

Instance warmup [Info](#)

300 seconds

☐ Disable scale in to create only a scale-out policy

16. Click Next until you reach Review. Then click on create auto scale group

The screenshot shows the 'Review' step of the 'Create Auto Scaling group' wizard in the AWS Management Console. A progress bar on the left indicates the current step. The main content area is divided into two sections: 'Step 1: Choose launch template or configuration' and 'Step 2: Choose instance launch options'. 'Step 1' includes 'Group details' (Auto Scaling group name: myasg1) and 'Launch template' (Launch template: mytemp1, Version: Latest, Description: template). 'Step 2' includes 'Network' (VPC: vpc-052c2b25aadcd07cd, Availability Zones and subnets table, Availability Zone distribution: Balanced best effort) and 'Instance type requirements' (This Auto Scaling group will adhere to the launch template.).

Review [Info](#)

Step 1: Choose launch template or configuration [Edit](#)

Group details

Auto Scaling group name
myasg1

Launch template

Launch template	Version	Description
mytemp1 lt-0ab54e0e141e6df5	Latest	template

Step 2: Choose instance launch options [Edit](#)

Network

VPC
[vpc-052c2b25aadcd07cd](#)

Availability Zones and subnets

Availability Zone	Subnet	Subnet CIDR range
us-east-1a	subnet-0dc025dd6af7da987	172.31.80.0/20

Availability Zone distribution
Balanced best effort

Instance type requirements
This Auto Scaling group will adhere to the launch template.

17. Auto Scaling group is successfully created

The screenshot shows the 'Auto Scaling groups' page in the AWS Management Console. A green notification banner at the top states 'myasg1, 1 Scaling policy created successfully'. Below the banner, there is a search bar and a table of Auto Scaling groups. The table has columns for Name, Launch template/configuration, Instances, Status, Desired capacity, Min, Max, Availability Zones, and Creation time. One group, 'myasg1', is listed with a status of 'Updating capacity...'. At the bottom, it says '0 Auto Scaling groups selected'.

Auto Scaling groups (1) [Info](#)

Last updated less than a minute ago [Launch configurations](#) [Launch templates](#) [Actions](#) [Create Auto Scaling group](#)

Search your Auto Scaling groups

☐	Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Availability Zones	Creation time
☐	myasg1	mytemp1 Version Latest	0	Updating capacity...	1	1	3	us-east-1a	Wed Sep 10 2025 08:55:14 GMT+05:30 (India St...)

0 Auto Scaling groups selected

18. EC2 instance for the autoscaling group is visible

The screenshot shows the AWS Management Console interface for the 'Instances' page. The left sidebar contains navigation links for EC2, Images, Elastic Block Store, Network & Security, and Load Balancing. The main content area displays a table of instances. The first instance, 'i-09a2b4ac62f4e3501', is in the 'Running' state. Below the table, there is a 'Select an instance' section.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic
	i-09a2b4ac62f4e3501	Running	t3.micro	initializing	View alarms +	us-east-1a	ec2-54-174-253-91.co...	54.174.253.91	-

19. Monitoring EC2 results:

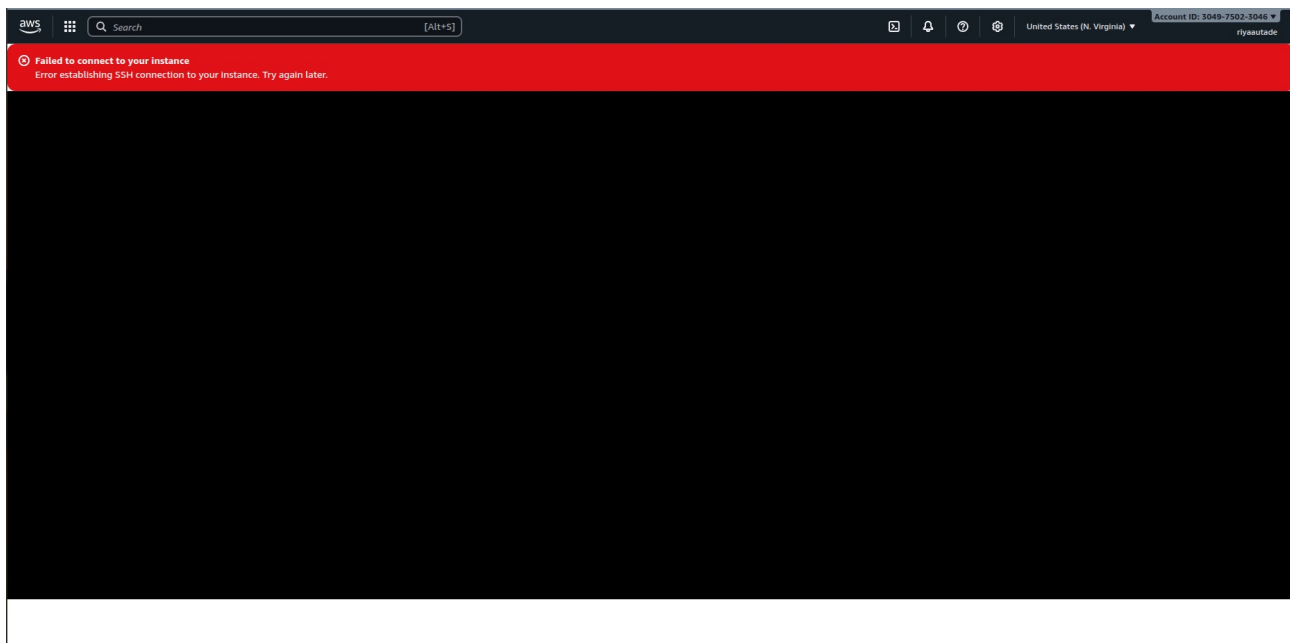
The screenshot shows the AWS Management Console interface for the 'Monitoring' page of the EC2 instance 'i-09a2b4ac62f4e3501'. The left sidebar contains navigation links for EC2, Images, Elastic Block Store, Network & Security, and Load Balancing. The main content area displays the 'Monitoring' tab, which shows various metrics for the instance. The metrics are displayed as line graphs over a time period of 1 hour. The metrics include CPU utilization (%), Network in (bytes), Network out (bytes), Network packets in (count), Network packets out (count), Metadata no token (count), CPU credit usage (count), and CPU credit balance (count).

Metric	Value
CPU utilization (%)	1.36
Network in (bytes)	360
Network out (bytes)	375
Network packets in (count)	4
Network packets out (count)	4.6
Metadata no token (count)	1
CPU credit usage (count)	0.301
CPU credit balance (count)	1.64

20. Connect to SSH

The screenshot shows the AWS Management Console interface for connecting to an EC2 instance. The breadcrumb navigation at the top indicates the path: EC2 > Instances > i-09a2b4ac62f4e3501 > Connect to instance. The main heading is 'Connect' with an 'Info' link. Below it, a sub-header states 'Connect to an instance using the browser-based client.' There are four tabs: 'EC2 Instance Connect' (selected), 'Session Manager', 'SSH client', and 'EC2 serial console'. Under the 'EC2 Instance Connect' tab, the 'Instance ID' is 'i-09a2b4ac62f4e3501'. The 'Connection type' section has two radio buttons: 'Connect using a Public IP' (selected) and 'Connect using a Private IP'. Below the 'Public IP' option, there are two sub-options: 'Public IPv4 address' (selected) with the value '54.174.253.91', and 'IPv6 address' which is currently empty. The 'Username' field contains 'ec2-user'. A note at the bottom states: 'Note: In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.' At the bottom right, there are 'Cancel' and 'Connect' buttons.

21. SSH Error on the system, therefore cannot verify:



21. Similar issue with public IP for Apache even though the Instance creation and security groups have been properly made and verified:

The screenshot displays the AWS Management Console for an EC2 instance. The left sidebar shows the navigation menu with categories like EC2, Images, Elastic Block Store, Network & Security, and Load Balancing. The main content area shows the 'Instance summary' for instance i-09a2b4ac62f4e3501. The instance is in a 'Running' state. The public IPv4 address is 54.174.253.91. The instance type is t3.micro. The AMI is ami-00ca32bbc84273381. The instance is managed by AWS.

Details	Status and alarms	Monitoring	Security	Networking	Storage	Tags
Instance details AMI ID ami-00ca32bbc84273381 AMI name al2023-ami-2023.8.20250818.0-kernel-6.1-x86_64 Stop protection	Monitoring disabled Allowed image - Launch time	Platform details Linux/UNIX Termination protection Disabled AMI location				

